
A Wasserstein Framework for Abnormal Batch Detection in Online SGD under Distributional Drift

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Abstract

Online stochastic gradient descent (OSGD) is widely used in modern machine learning and AI systems for learning from streaming data. However, abnormal data batches caused by corruption, adversarial manipulation, or annotation errors can destabilize the optimization process and degrade model reliability. Detecting such abnormal batches is particularly challenging under natural distributional drift, where genuine anomalies are entangled with normal temporal variation. Existing methods suffer from major limitations: gradient- and loss-based methods are sensitive to stochastic noise, robust-statistics methods rely on restrictive assumptions, and two-sample testing methods collapse historical batches into a single reference distribution, failing to capture the evolution of streaming data. We propose a new framework for abnormal batch detection in OSGD under natural distributional drift. Unlike existing approaches that operate on Euclidean data points, our method treats streaming batches as distribution-valued random objects in Wasserstein space. We employ Wasserstein Spatial Depth to quantify batch centrality and further transform the depth values into empirical p-values, yielding a fully nonparametric and distribution-adaptive measure of abnormality. The resulting procedure detects and excludes abnormal batches during OSGD updates, thereby stabilizing online optimization. Theoretical analysis, numerical experiments and real-data studies demonstrate that the proposed method consistently outperforms several classical anomaly detection approaches under natural distributional drifts.

1 Introduction

In modern machine learning and AI systems, online stochastic gradient descent (OSGD) is widely used to continuously update models from streaming or sequentially arriving data batches. Let $\mathbf{X}^t$, $t = 1, \dots, T$ denote the sequence of data batches, the classical OSGD update takes the form

$$\theta_{t+1} = \theta_t - \eta_t \nabla \ell(\theta_t; \mathbf{X}^t), \quad (1)$$

where $\eta_t > 0$ is the learning rate and $\ell(\cdot; \cdot)$ is the loss function.

In dynamic environments, certain batches may be abnormal due to corrupted measurements, abrupt distributional shifts, adversarial manipulation, sensor failures, annotation errors, or rare unexpected events. Detecting such abnormal batches is critical for trustworthy AI, since even a small number of anomalous updates can destabilize the optimization trajectory, degrade predictive performance, induce hallucination or catastrophic forgetting, and introduce harmful biases into deployed systems. In safety-sensitive applications such as healthcare, autonomous systems, and foundation model training, abnormal batches can propagate errors throughout OSGD and compromise robustness, reliability, fairness, and interpretability. Consequently, detecting abnormal batches during OSGD is essential for reliable learning under evolving and contaminated data streams.

35 A fundamental challenge arises from the presence of natural *distributional drift*. In practice, streaming
 36 batches are rarely generated from a fixed distribution; instead, the underlying data-generating mecha-
 37 nism evolves over time due to environmental changes, equipment degradation, human behavioral
 38 variability, or preprocessing effects. Under such drift, it is fundamentally challenging to disentangle
 39 genuinely abnormal batches from natural distributional variation.

40 1.1 Related works

41 Existing approaches for anomaly detection in online stochastic optimization fall within four categories.
 42 The first class consists of gradient-based approaches, which compare the current gradient with
 43 historical gradients [Blanchard et al., 2017, Yin et al., 2018]. The second class includes loss-based
 44 methods that monitor jumps in the loss function or deviations from moving averages [Gama et al.,
 45 2004, Stucchi et al., 2022]. The third class focuses on distribution-shift detection by comparing the
 46 empirical distribution of the current batch with a reference distribution estimated from historical data
 47 [Gao et al., 2020, Ramdas et al., 2017]. Classical approaches based on sequential analysis, robust
 48 statistics, and window smoothing have also been widely studied [Laxhammar and Falkman, 2013,
 49 Baena-Garcia et al., 2006, Blanchard et al., 2017, Gama et al., 2014, Hinder et al., 2024].

50 Although these works substantially advance the field, they bear some limitations. Gradient- and
 51 loss-based methods rely on low-dimensional summary statistics and are sensitive to stochastic
 52 optimization noise and training dynamics, making it difficult to distinguish true abnormalities from
 53 natural randomness or gradual drift. Methods grounded in robust statistics or sequential analysis
 54 rely on restrictive assumptions such as sparse contamination, stationarity, or *i.i.d.* batches, which
 55 are unrealistic in nonstationary online learning environments. Existing distribution-shift detection
 56 methods typically operate at the level of individual data points by pooling historical regular batches
 57 into a single reference dataset and comparing the current batch against it via two-sample testing, KL
 58 divergence, or Wasserstein distance. Such pooling may obscure mild but systematic drifts among
 59 regular batches and therefore reduce sensitivity to genuinely abnormal batches.

60 The challenge becomes particularly acute under natural distributional drift. As a toy example, two
 61 regular batches follow $\text{Unif}[-0.5, 0]$ and $\text{Unif}[0, 0.5]$, respectively, while an abnormal batch follows
 62 $\text{Unif}[-0.5, 0.5]$, which coincides with the mixture of the two regular batches. In such cases, methods
 63 based on pooled regular/normal batches will fail to detect the abnormal batch. This illustrates the
 64 necessity of modeling variability at the batch level rather than collapsing all historical regular batches
 65 into a single reference distribution.

66 1.2 Our contribution

67 We propose a framework for detecting and excluding abnormal data batches in OSGD under natural
 68 distributional drift, thereby stabilizes online optimization. Unlike existing methods that operate on
 69 individual data points, our approach treats streaming batches as distribution-valued random objects in
 70 Wasserstein space. This distributional viewpoint enables the method to account for the natural batch
 71 evolution.

72 2 Methodology

73 Throughout this section, let $\mathcal{P}_2(\mathbb{R}^d)$ denote the Wasserstein-2 space of probability distributions on
 74 $\mathbb{R}^d$. Let P_t be the underlying distribution generating the data points in the t -th batch $\mathbf{X}^t$, *i.e.*, the rows
 75 of $\mathbf{X}^t$ are *i.i.d.* samples from P_t . Hereinafter, we denote the i -th data point within the t -th batch (the
 76 i -th row of $\mathbf{X}^t$) as X_i^t . To accommodate realistic streaming settings, we allow P_t to evolve gradually
 77 over time and regard each P_t as a random object taking values in the Wasserstein space $\mathcal{P}_2(\mathbb{R}^d)$. Let
 78 $\mathbf{P}$ denote a distribution on $\mathcal{P}_2(\mathbb{R}^d)$ (a distribution of probability distributions) that characterizes the
 79 natural variability of regular data batches. Let $\{P_t\}_{t \in \mathcal{I}_r}$ be a collection of regular distributions, called
 80 the *reference collection*, where $|\mathcal{I}_r| = R$, and assume that these distributions are *i.i.d.* draws from $\mathbf{P}$.

81 We model streaming data with natural distributional drift through the two-stage sampling framework

$$P_t \stackrel{iid}{\sim} \mathbf{P} \text{ for } t \in \mathcal{I}_r, \quad X_i^t \stackrel{iid}{\sim} P_t, \quad \hat{P}_t = \frac{1}{m} \sum_{i=1}^m \delta_{X_i^t}, \quad \hat{\mathbf{P}}_{R,m} = \frac{1}{R} \sum_{t \in \mathcal{I}_r} \delta_{\hat{P}_t}. \quad (2)$$

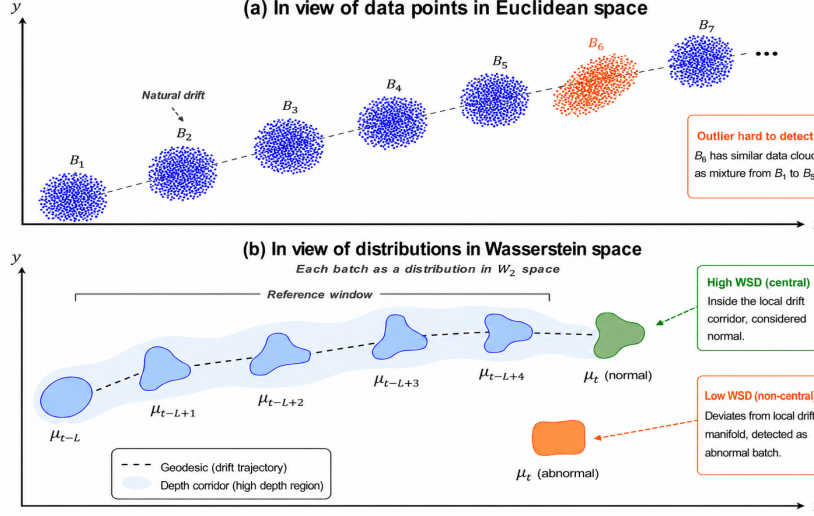


Figure 1: The advantage of treating data batches as distributions and analyzing them in Wasserstein space.

That is, one first samples distributions $\{P_t, t \in \mathcal{I}_r\}$ from $\mathbf{P}$ and then samples data points from each P_t . For simplicity, we assume a common batch size m , although varying batch sizes are also allowed. Under this framework, detecting abnormal batches becomes the problem of identifying outlier distributions among $\hat{P}_1, \dots, \hat{P}_T$. Thus, the analysis is performed at the level of distributions rather than Euclidean data points.

The proposed framework has three main advantages: (i) it explicitly models natural distributional drift through variability across batches; (ii) it preserves the batch-evolution structure instead of collapsing all regular batches into a single reference dataset; (iii) it often provides a low-dimensional distributional representation that mitigates the curse of dimensionality in high-dimensional data analysis.

2.1 Empirical p-value based on Wasserstein spatial depth

To detect abnormal distributions, we employ *Wasserstein Spatial Depth* proposed in [Bachoc et al., 2024], which provides a notion of statistical depth “rank” distributions in Wasserstein space.

Definition 2.1. The Wasserstein Spatial Depth (WSD) of any distribution Q with respect to $\mathbf{P}$ is defined as

$$\text{WSD}(Q; \mathbf{P}) = 1 - \left\| \mathbb{E}_{P \sim \mathbf{P}} \left[\frac{Id - T_{Q,P}}{W_2(Q, P)} \right] \right\|_{L^2(Q)}, \quad (3)$$

where $T_{Q,P}$ is the optimal transport map from Q to P and Id is the identity map.

Intuitively, WSD measures the centrality of a distribution within the population $\mathbf{P}$ under Wasserstein geometry. Distributions with small WSD values are less representative of the population and are therefore more likely to be outliers.

Under the framework in (2), the abnormality of the t -th batch is quantified by

$$\text{WSD}(\hat{P}_t; \hat{\mathbf{P}}_{R,m}) = 1 - \frac{1}{m} \sum_{i=1}^m \left\| \frac{1}{R} \sum_{j \in \mathcal{I}_r, j \neq t} \left[\frac{X_i^t - T_{\hat{P}_t, \hat{P}_j}(X_i^t)}{W_2(\hat{P}_t, \hat{P}_j)} \right] \right\|. \quad (4)$$

Treating each batch as an empirical distribution $\hat{P}_t$ and the reference collection as an empirical distribution of distributions $\hat{\mathbf{P}}_{R,m}$ naturally aligns WSD with our objective of abnormal batch detection. Regular batches tend to have large WSD values, whereas abnormal batches have small WSD values.

A direct strategy for stabilizing OSGD is to modulate the contribution of each batch according to its WSD value. However, the distribution of $\text{WSD}(P; \mathbf{P}) : P \sim \mathbf{P}$ depends on the unknown

population $\mathbf{P}$, making raw WSD values difficult to interpret across different streaming environments. In particular, there is generally no universal cutoff value for WSD with a meaningful statistical significance level.

To obtain a normalized and distribution-adaptive measure of abnormality, we transform WSD into the score

$$w_t = \mathbb{P}_{P \sim \mathbf{P}}\{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(P_t; \mathbf{P})\}, \quad (5)$$

which is the cumulative distribution function (CDF) of $\text{WSD}(P; \mathbf{P})$ evaluated at $\text{WSD}(P_t; \mathbf{P})$. Smaller values of w_t indicate that the current batch is less representative of the regular population and therefore more likely to be abnormal. Its empirical counterpart is

$$\hat{w}_t = \frac{1}{R} \sum_{j \in \mathcal{I}_r} \mathbf{1}\{\text{WSD}(\hat{P}_j; \hat{\mathbf{P}}_{R,m}) \leq \text{WSD}(\hat{P}_t; \hat{\mathbf{P}}_{R,m})\}. \quad (6)$$

The quantity $\hat{w}_t$ can be interpreted as an empirical p-value. If $P_t \sim \mathbf{P}$, the rank of $\text{WSD}(\hat{P}_t; \hat{\mathbf{P}}_{R,m})$ among the reference WSD values $\{\text{WSD}(\hat{P}_j; \hat{\mathbf{P}}_{R,m}), j \in \mathcal{I}_r\}$ is approximately uniform. Consequently, $\hat{w}_t$ admits a unified probabilistic interpretation across heterogeneous data streams: small values indicate that the current batch is less normal than most regular batches and is therefore potentially anomalous. Moreover, under regularity conditions, $\hat{w}_t$ approximately follows a $\text{Unif}(0, 1)$ distribution, allowing statistically meaningful cutoff values to be chosen naturally.

Remark The proposed method elevates the analysis from Euclidean data points to probability distributions in Wasserstein space, enabling abnormality detection at the batch level. Unlike existing approaches that pool regular batches into a single reference distribution, our framework treats streaming batches as distribution-valued random objects, preserving batch-evolution structure and capturing higher-order distributional changes beyond mean or covariance shifts. By combining Wasserstein Spatial Depth with an empirical p-value transformation, the method provides a fully nonparametric, geometry-aware, and distribution-adaptive measure of abnormality with a unified probabilistic interpretation across heterogeneous data streams. Some intuition is provided in Figure 1.

2.2 Online SGD with WSD-based anomaly filtering

The key idea is to exclude abnormal batches during OSGD updates by treating any batch with $\hat{w}_t < \alpha$ as an outlier. The proposed update rule is

$$\theta_{t+1} = \theta_t - \mathbf{1}\{\hat{w}_t \geq \alpha\} \eta_t \nabla \ell(\theta_t; \mathbf{X}^t). \quad (7)$$

This mechanism protects the optimization trajectory from contaminated batches while preserving the standard SGD dynamics on regular data. The full procedure is summarized in Algorithm 1 in Appendix A.

3 Theoretical results

We derive some theoretical properties for the WSD-based p-value defined in (5), while proofs are deferred to Appendix B. For better delivery of the theory, define

$$R(Q; \mathbf{P}) = \mathbb{P}_{P \sim \mathbf{P}}\{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})\}. \quad (8)$$

It is clear that $w_t = R(P_t; \mathbf{P})$.

Proposition 3.1. Assume $P_t \sim \mathbf{P}$. If $\mathbf{P}$ is supported on $\mathcal{P}_2^{a.c.}(\mathbb{R}^d)$ and such that for a fixed constant $C > 0$

$$\mathbb{P}_{P \sim \mathbf{P}}\{t_1 < \text{WSD}(P; \mathbf{P}) \leq t_2\} \leq C|t_2 - t_1|, \quad \forall t_1, t_2 \in [0, 1].$$

Then it holds (i) $R(P_t; \mathbf{P}) \sim \text{Unif}[0, 1]$; (ii) $R(Q; \mathbf{P})$ is transformation invariant in the sense that for any isometry $F : \mathcal{P}(\mathbb{R}^d) \rightarrow \mathcal{P}(\mathbb{R}^d)$, we have $R(F(Q), F(\mathbf{P})) = R(Q, \mathbf{P})$, where $F(\mathbf{P})$ denotes the distribution of $F(P)$ for $P \sim \mathbf{P}$.

The idea can be extended to address the problem of testing whether two streams of data batches are homogeneous. Towards this end, we introduce a quantity

$$D(\mathbf{Q}, \mathbf{P}) = \mathbb{P}_{Q \sim \mathbf{Q}, P \sim \mathbf{P}}\{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})\}, \quad (9)$$

147 which is the expectation of $R(Q; \mathbf{P})$ under $Q \sim \mathbf{Q}$. It is easy to see that $D(\mathbf{Q}, \mathbf{P}) = 1/2$ when
 148 $\mathbf{Q} = \mathbf{P}$, and it is also transformation invariant. The next Theorem gives the asymptotic distribution
 149 of $D(\mathbf{Q}_n, \mathbf{P}_n)$, where $\mathbf{Q}_n, \mathbf{P}_n$ are empirical counterparts of $\mathbf{Q}, \mathbf{P}$, respectively. We can construct a
 150 two-stream test based on Theorem 3.2.

151 **Theorem 3.2.** Let $P_i \stackrel{i.i.d.}{\sim} \mathbf{P}$, $i = 1, \dots, n$ and $Q_k \stackrel{i.i.d.}{\sim} \mathbf{Q}$, $k = 1, \dots, n$ be independent samples.
 152 Suppose $\mathbf{P}, \mathbf{Q}$ are supported on $\mathcal{P}_2^{a.c.}(\mathbb{R}^d)$, and such that for a fixed constant $C' > 0$,

$$\mathbb{P}_{Q \sim \mathbf{Q}}\{t_1 < \text{WSD}(Q; \mathbf{P}) \leq t_2\} \leq C'|t_2 - t_1|, \quad \forall t_1, t_2 \in [0, 1] \quad (10)$$

$$\sup_{A \in \text{supp}(\mathbf{P}) \cup \text{supp}(\mathbf{Q})} |\text{WSD}(A; \mathbf{P}_n) - \text{WSD}(A; \mathbf{P})| = o(1) \text{ as } n \rightarrow \infty \quad (11)$$

$$\mathbb{E} \left[\sup_{A \in \text{supp}(\mathbf{P}) \cup \text{supp}(\mathbf{Q})} |\text{WSD}(A; \mathbf{P}_n) - \text{WSD}(A; \mathbf{P})| \right] = O(n^{-1/2}) \quad (12)$$

$$\int \int I_{\text{WSD}(P; \mathbf{P}_n) = \text{WSD}(Q; \mathbf{P}_n)} d\mathbf{P} d\mathbf{Q} = o(n^{-1/2}). \quad (13)$$

155 Then, it holds

$$\sqrt{\frac{n}{(\sigma_{\mathbf{P}\mathbf{Q}}^2 + \sigma_{\mathbf{Q}\mathbf{P}}^2)}} \left(D(\mathbf{Q}_n, \mathbf{P}_n) - D(\mathbf{Q}, \mathbf{P}) \right) \xrightarrow{d} N(0, 1) \text{ as } n \rightarrow \infty,$$

156 where

$$\begin{aligned} \sigma_{\mathbf{Q}\mathbf{P}}^2 &= \int \left(\mathbb{P}_{Q \sim \mathbf{Q}}\{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})\} \right)^2 d\mathbf{P} - D^2(\mathbf{Q}, \mathbf{P}), \\ \sigma_{\mathbf{P}\mathbf{Q}}^2 &= \int \left(\mathbb{P}_{P \sim \mathbf{P}}\{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})\} \right)^2 d\mathbf{Q} - D^2(\mathbf{Q}, \mathbf{P}). \end{aligned}$$

157 Proposition 3.1 justifies that w_t is a p-value. If $\hat{w}_t$ is very small, it is unlikely that $P_t \sim \mathbf{P}$, indicating
 158 that P_t is an anomalous batch. When assessing whether two data streams are homogeneous or whether
 159 a new data stream preserves the quality or structural characteristics of a reference stream, we evaluate
 160 $D(\mathbf{Q}_n, \mathbf{P}_n)$. Guided by Theorem 3.2, if $D(\mathbf{Q}_n, \mathbf{P}_n)$ deviates substantially from $1/2$, it is unlikely
 161 that these two data streams are from the same population.

162 4 Numerical simulation

163 We compare the proposed method with the common approaches based on two-sample testing, and
 164 further evaluate its performance in online regression settings. In practice, the effective batches are
 165 the gradient batches $\{\nabla \ell(\theta; X_i^t)\}_{i=1}^m$, as parameter updates depend only on gradients. Accordingly,
 166 in our experiments, $\hat{P}_t$ denotes the empirical distribution of the t -th batch of gradients. For fair
 167 comparison, we compute WSD using gradient batches evaluated at a fixed anchor parameter θ_0 ,
 168 namely $\{\nabla \ell(\theta_0; X_i^t)\}_{i=1}^m$, while the actual parameter updates still use the instantaneous gradients
 169 $\nabla \ell(\theta_t; X_i^t)$. Details about data synthesis and computation are deferred to Appendix C.

170 4.1 Comparing to two-sample testing

171 A common approach for anomaly detection in data streams is to perform two-sample testing between
 172 the current batch and a reference distribution formed by pooling all regular batches. We compare
 173 the proposed method with three classical two-sample tests: the Kolmogorov–Smirnov (KS) test [An,
 174 1933, Smirnov, 1948], energy distance [Székely and Rizzo, 2013], and Maximum Mean Discrepancy
 175 (MMD) [Gretton et al., 2012]. Regular batches are generated from $\mathbf{P}$, while abnormal batches are
 176 generated from $\mathbf{Q}$, with each batch containing 50 data points. We consider two cases below. In each
 177 case, we follow this procedure: generate 100 reference batches from $\mathbf{P}$ and simulate a data stream
 178 consisting of 20 abnormal batches from $\mathbf{Q}$ and 20 regular batches from $\mathbf{P}$; for each incoming batch,
 179 perform two-sample testing against the pooled reference dataset containing 100×50 data points, and
 180 compute the proposed WSD-based empirical p-values.

181 **Case 1:**

$$\begin{aligned} \mathbf{P} &= \{ \text{Unif}[c-1, c+1], c \sim \text{Triangle}(-1, 0, 1) \}; \\ \mathbf{Q} &= \{ \text{discrete distribution supported on equidistant 200 points within } [-2, 2] \\ &\quad \text{with PMF equal to the histogram of } N(0, 1) \}. \end{aligned}$$

182 As shown in Figure 2, each abnormal batch exhibits a data cloud similar to that of the pooled regular
183 batches. Consequently, classical two-sample tests fail to detect the outlier batches, whereas the
184 WSD-based p-values successfully identify them, as demonstrated in Table 1.

185 **Case 2:**

$$\begin{aligned} \mathbf{P} &= \{ \text{Unif}[\text{a two-dimensional annulus with fixed area}], \text{annulus inner radius } r_0 \sim \text{Unif}[0.02, 1] \}; \\ \mathbf{Q} &= \{ \text{random data points } (r \cos \omega, r \sin \omega) \text{ with } \omega \sim \text{Unif}[(0, 2\pi], r \sim \sqrt{2} \text{Beta}(4, 3) \}. \end{aligned}$$

186 As shown in Figure 2, each abnormal batch has a similar data cloud to that of the pooled regular
187 batches. Consequently, classical two-sample tests fail to detect the outlier batches, while WSD-based
188 p-values work well (shown in Table 1). Since KS test only applies to one-dimensional data, we do
189 not report it for Case 2.

Table 1: Comparing to two-sample tests between two sets of Euclidean data points. The reported numbers are p-values averaged over 20 independent replications (mean $\pm$ standard deviation).

Setting		KS	Energy	MMD	WSD-based p -value
Case 1	abnormal	0.264 ± 0.237	0.163 ± 0.185	0.148 ± 0.182	$(3.91 \pm 12.10) \times 10^{-41}$
	regular	0.075 ± 0.117	0.053 ± 0.100	0.062 ± 0.110	0.082 ± 0.114
Case 2	abnormal	N/A	0.362 ± 0.241	0.317 ± 0.210	$(4.53 \pm 18.23) \times 10^{-5}$
	regular	N/A	0.128 ± 0.101	0.079 ± 0.061	0.237 ± 0.248

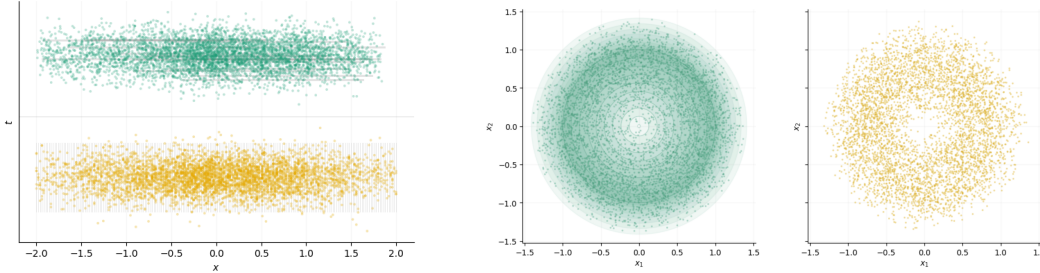


Figure 2: Similarity between abnormal batches and the pooled reference batches. Green cloud represents the superposition of 20 reference batches and the yellow cloud represents the superposition of 20 abnormal batches. Left: Case 1; Middle and Right: Case 2.

190 4.2 Anomaly detection in online regression

191 We then evaluate the performance of WSD based p-value for anomaly detection in online regression
192 setting. The t -th data batch contains $\{(y_i^t, x_i^t)\}_{i=1}^m$ where $y_i^t \in \mathbb{R}$, $x_i^t \in \mathbb{R}^d$. The data points within
193 each batch is generated by a linear regression model

$$y_i^t = \langle x_i^t, \theta^t \rangle + \varepsilon_i^t, \quad x_i^t \sim N(0, \mathbf{I}_{d \times d}), \quad \varepsilon_i^t \sim N(0, 1). \quad (14)$$

194 The $\theta^t \in \mathbb{R}^d$ slowly evolves with t via the following mechanism

$$\theta^{t+1} = \theta^t + N(0, 0.1^2 \mathbf{I}_{d \times d}), \quad \text{coordinates of } \theta^0 \text{ are iid drawn from } \text{Unif}[-2, -1) \cup (0.5, 2.5].$$

195 We introduce abnormal/outlier batches by perturbing the coordinates of θ^t . For an anomalous t ,

$$y_i^t = \langle x_i^t, \theta^t + u_i^t \rangle + \varepsilon_i^t, \quad u_i^t \stackrel{\text{iid}}{\sim} N(0, 0.6^2 \mathbf{I}_{d \times d}). \quad (15)$$

During implementation, we set $d = 5$, $m = 60$, and $T = 120$. The first 25 batches are used as a burn-in phase to warm-start $\hat{\theta}$. The remaining 95 batches are monitored for abnormality, among which 14 are truly anomalous. The reference collection is chosen as the previous 20 batches using a sliding window scheme; that is, $\{\mathbf{X}^{t-20}, \mathbf{X}^{t-19}, \dots, \mathbf{X}^{t-1}\}$ is used to determine whether $\mathbf{X}^t$ is an anomalous batch.

Figure 3 (left panel) displays the trajectory of WSD-based p-values along the data stream. The abnormal batches generally exhibit smaller p-values than the regular batches, consistent with the right panel. The right panel further shows that the WSD-based p-values of regular batches approximately follow a $\text{Unif}(0, 1)$ distribution.

We compare WSD-based p-values to four classical approaches for anomaly detection in OSGD: gradient norm [Huang et al., 2021], Mean-grad L_2 [Rabanser et al., 2019], Hotelling T^2 [Hotelling et al., 1931], and MMD [Gretton et al., 2012]. Receiver Operating Characteristic (ROC) curves are used to evaluate the five anomaly detection methods by illustrating the tradeoff between detection power and false alarm rate. As shown in Figure 4, the ROC curve of the WSD-based p-value lies mostly above those of the other methods, indicating superior detection performance. Consistently, Table 2 shows that the WSD-based p-value achieves the largest Area Under the ROC Curve (AUC) among all five methods.

We also observe improved estimation accuracy in OSGD with anomaly filtering (dynamic MSE=0.026) compared to the vanilla OSGD using all data batches (dynamic MSE=0.11).

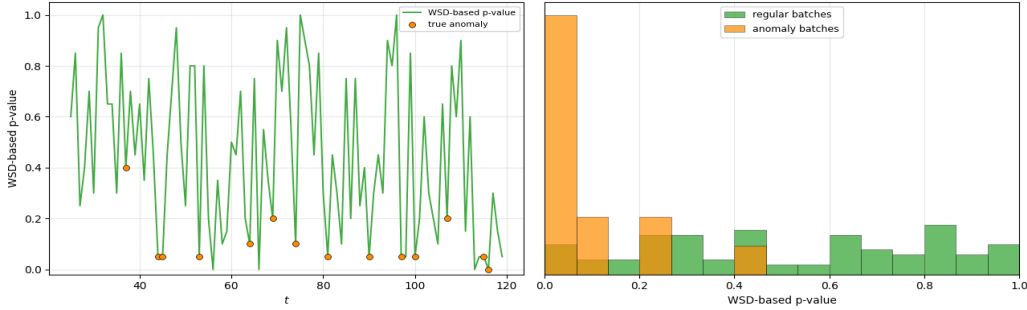


Figure 3: Anomaly detection using WSD-based p-values in online regression. Left panel: trajectory of the empirical p-values; orange dots mark the true anomalies. Right panel: empirical distribution of the WSD-based p-values of regular batches vs. abnormal batches.

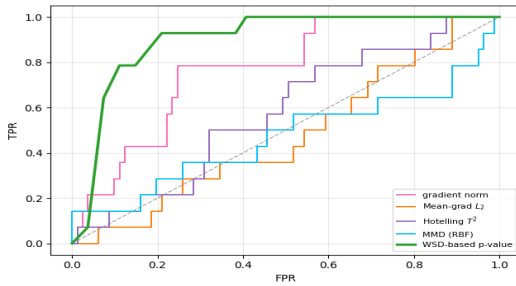


Figure 4: ROC curves of five methods in the online regression stream.

Table 2: AUCs of five methods (mean $\pm$ std) in the online regression stream, averaged over 20 independent replications. A larger AUC indicates a more powerful detector.

Anomaly detector	AUC
WSD-based p -value	0.848 ± 0.053
Gradient norm	0.761 ± 0.074
Hotelling T^2	0.620 ± 0.076
MMD (RBF)	0.591 ± 0.089
Mean-grad L_2	0.523 ± 0.092

5 Real data analysis

We apply the proposed method to the MNIST dataset of handwritten digits. The task is to train a CNN classifier to distinguish the images of ten digits using online SGD with streaming data batches, mimicking the continuous arrival of new images in real-world settings. Regular batches contain correctly labeled images and exhibit natural distributional drift through gradual changes in digit proportions. Specifically, the class proportions evolve along the interpolating trajectory

221 $\pi_t = \frac{T-1-t}{T-1}\pi_0 + \frac{t}{T-1}\pi_1$ for $t = 0, 1, \dots, T-1$, where π_0, π_1 are independently drawn from
 222 Dirichlet(0.4) of order 10. Abnormal batches contain 10% mislabeled images. We perform WSD-
 223 based outlier detection over $T = 120$ batches, among which 12 are abnormal. Each batch contains
 224 $m = 60$ images, and a batch is declared abnormal if its WSD-based p-value is below $\alpha = 0.05$. The
 225 experiment details are deferred to Appendix D.

226 Figure 5 shows that abnormal batches have smaller WSD-based p-values than regular batches, while
 227 the p-values of regular batches approximately follow a $\text{Unif}(0, 1)$ distribution. Figure 6 and Table
 228 3 further demonstrate that the proposed method outperforms the four classical two-sample tests
 229 for anomaly detection in OSGD. The classifier trained by OSGD with anomaly filtering has better
 230 accuracy (mean accuracy = 0.82) compared to that trained by vanilla OSGD using all data batches
 231 (mean accuracy = 0.68).

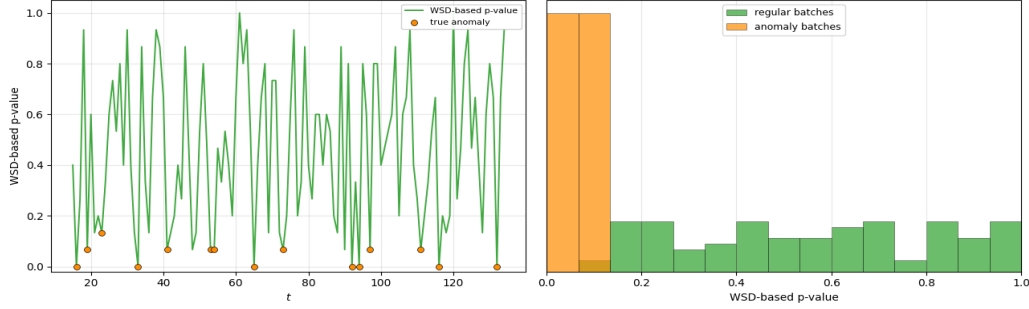


Figure 5: Anomaly detection using WSD-based p-values in streaming batches of MNIST images. Left panel: trajectory of the empirical p-values; orange dots mark the true anomalies. Right panel: empirical distribution of the WSD-based p-values of regular batches vs. abnormal batches.

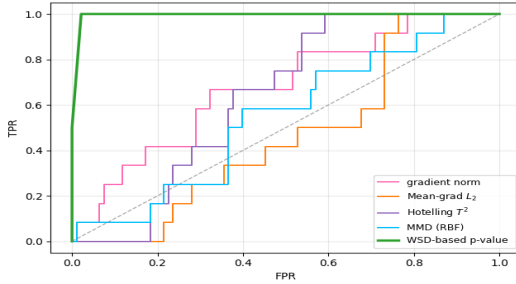


Figure 6: ROC curves of five methods.

Table 3: AUCs of five methods(mean $\pm$ std): a larger AUC indicates a more powerful detector.

Anomaly detector	AUC
WSD-based p -value	0.957 ± 0.038
Hotelling T^2	0.719 ± 0.070
Gradient norm	0.700 ± 0.044
Mean-grad L_2	0.629 ± 0.112
MMD (RBF)	0.621 ± 0.061

232 6 Discussion

233 The proposed methodology introduces a new distributional framework for abnormal batch detection in
 234 online learning under natural distributional drifts. It treats streaming batches as random distributions,
 235 which captures the variation at the batch level and preserves the intrinsic evolution structure of stream-
 236 ing data. By combining WSD with an empirical p-value transformation, this framework provides
 237 a fully nonparametric, geometry-aware, and distribution-adaptive measure of abnormality with a
 238 unified probabilistic interpretation. Beyond anomaly filtering in OSGD, the proposed framework is
 239 broadly applicable for studying distributional drifts in multi-batch settings, particularly in trustworthy
 240 AI, continual learning, and foundation-model training under evolving data streams. Several directions
 241 warrant further investigation, including dynamic-regret analysis under nonstationary environments,
 242 statistical guarantees under gradual drift and adversarial contamination, *etc.* Extending the framework
 243 to structured data objects such as graphs, images, manifolds, and functional data is another promising
 244 direction.

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291 A Algorithm

This is the procedure of online SGD with WSD-based anomaly filtering.

Algorithm 1 Online Stochastic Gradient Descent with Anomaly Detection

```

1: Input: a collection of reference batches/distributions  $\{\hat{P}_i\}_{i \in \mathcal{I}_r}$ , streaming data batches  $\{\mathbf{X}^t\}_{t=1}^T$ ,
   gradient formula  $\nabla \ell(\theta; \cdot)$ , level  $\alpha$ 
2: for  $t = 1, \dots, T$  do
3:   compute  $\text{WSD}(\hat{P}_t; \hat{\mathbf{P}}_{R,m})$  as in equation (4) and the empirical p-value  $\hat{w}_t$  as in (6).
4:   if  $\hat{w}_t < \alpha$  then
5:     flag batch  $t$  as anomalous and discard this batch; skip
6:   else
7:     update  $\theta_t$  with the new gradient descent  $\nabla \ell(\theta_t; \mathbf{X}^t)$ .
8:   end if
9: end for

```

292

293 Evaluating WSD is a central part when implementing our method. The key step of evaluating WSD
 294 is to compute the empirical optimal transport (OT) map between two distributions. For two empirical
 295 measures

$$\hat{P} = \frac{1}{B} \sum_{i=1}^B \delta_{g_i}, \quad \hat{Q} = \frac{1}{B} \sum_{j=1}^B \delta_{h_j},$$

296 the ground cost is the squared Euclidean cost

$$C_{ij} = \|g_i - h_j\|^2.$$

297 The OT plan is computed by entropic Sinkhorn with the log-domain stabilization. The entropic
 298 regularization is set adaptively as

$$\varepsilon = 0.01 \cdot \text{median}(C),$$

299 which keeps the regularization at a fixed fraction of the cost-matrix scale and avoids the numerical
 300 issues that arise when the same fixed ε is reused across streams of different gradient dimensionalities.
 301 The iteration budget is 200.

302 B Main proofs

303 We give the proofs for Proposition 3.1 and Theorem 3.2.

304 B.1 Proofs for Proposition 3.1

305 *Proof.* (i) Define $Z_P = \text{WSD}(P; \mathbf{P})$. By Theorem 5.1 in [Bachoc et al., 2024] and the assumption

$$\mathbb{P}_{P \sim \mathbf{P}}\{t_1 < \text{WSD}(P; \mathbf{P}) \leq t_2\} \leq C|t_2 - t_1|, \quad \forall t_1, t_2 \in [0, 1],$$

306 we obtain that Z_P is a continuous random variable taking values on $[0, 1]$. Let z be any realization of
 307 Z_P , then we have

$$\mathbb{P}\{R(P, \mathbf{P}) \leq t\} = \mathbb{P}\{\mathbb{P}_{Z_P}\{Z_P \leq z\} \leq t\} = \mathbb{P}\{F_{Z_P}(z) \leq t\} = t,$$

308 where $F_{Z_P}(\cdot)$ is the cumulative distribution function of Z_P .

309 (ii) directly follows from the transform invariance property of $\text{WSD}(\cdot; \cdot)$ in Theorem 5.1 in [Bachoc
 310 et al., 2024]. $\square$

311 B.2 Proofs for Theorem 3.2

312 *Proof.* First, we claim that

$$D(\mathbf{Q}_n, \mathbf{P}_n) - D(\mathbf{Q}_n, \mathbf{P}) = \int \int I_{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})} d\mathbf{Q} d(\mathbf{P}_n - \mathbf{P}) + o_p(n^{-1/2}). \quad (16)$$

313 To show this claim, we derive as follows

$$\begin{aligned}
& D(\mathbf{Q}_n, \mathbf{P}_n) - D(\mathbf{Q}_n, \mathbf{P}) \\
&= \int \int I_{\text{WSD}(P; \mathbf{P}_n) \leq \text{WSD}(Q; \mathbf{P}_n)} d\mathbf{P}_n d\mathbf{Q}_n - \int \int I_{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})} d\mathbf{P} d\mathbf{Q}_n \\
&= \int \int \left(I_{\text{WSD}(P; \mathbf{P}_n) \leq \text{WSD}(Q; \mathbf{P}_n)} - I_{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})} \right) d\mathbf{P}_n d\mathbf{Q}_n \\
&+ \int \int I_{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})} d(\mathbf{Q}_n - \mathbf{Q}) d(\mathbf{P}_n - \mathbf{P}) \\
&+ \int \int I_{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})} d\mathbf{Q} d(\mathbf{P}_n - \mathbf{P}).
\end{aligned}$$

314 We will denote the above three terms as T_1, T_2, T_3 . By the central limit theory for empirical
315 distribution process and Theorem 6.1 in [Bachoc et al., 2024], we have $\sqrt{n} T_2 = o_p(1)$. Next, we
316 decompose T_1 into

$$\begin{aligned}
T_1 &= \int \int \left(I_{\text{WSD}(P; \mathbf{P}_n) \leq \text{WSD}(Q; \mathbf{P}_n)} - I_{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})} \right) d\mathbf{P}_n d(\mathbf{Q}_n - \mathbf{Q}) \\
&+ \int \int \left(I_{\text{WSD}(P; \mathbf{P}_n) \leq \text{WSD}(Q; \mathbf{P}_n)} - I_{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})} \right) d\mathbf{P} d\mathbf{Q} \\
&+ \int \int \left(I_{\text{WSD}(P; \mathbf{P}_n) \leq \text{WSD}(Q; \mathbf{P}_n)} - I_{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})} \right) d(\mathbf{P}_n - \mathbf{P}) d\mathbf{Q}.
\end{aligned}$$

317 Denote the above three terms as T_{11}, T_{12} , and T_{13} . By assumptions (10) and (12), we have

$$\begin{aligned}
\mathbb{E}[T_{11}^2] &\leq \mathbb{E} \left[\int \left[\int \left(I_{\text{WSD}(P; \mathbf{P}_n) \leq \text{WSD}(Q; \mathbf{P}_n)} - I_{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})} \right) d(\mathbf{Q}_n - \mathbf{Q}) \right]^2 d\mathbf{P}_n \right] \\
&= \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n \left[\int \left(I_{\text{WSD}(P_i; \mathbf{P}_n) \leq \text{WSD}(Q; \mathbf{P}_n)} - I_{\text{WSD}(P_i; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})} \right) d(\mathbf{Q}_n - \mathbf{Q}) \right]^2 \right] \\
&= \mathbb{E} \left[\int \left(I_{\text{WSD}(P; \mathbf{P}_n) \leq \text{WSD}(Q; \mathbf{P}_n)} - I_{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})} \right) d(\mathbf{Q}_n - \mathbf{Q}) \right]^2 \\
&\leq \frac{1}{n} \mathbb{E} \left[\mathbb{E}_Q \left[\left(I_{\text{WSD}(P; \mathbf{P}_n) \leq \text{WSD}(Q; \mathbf{P}_n)} - I_{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})} \right)^2 \middle| P_1, \dots, P_n \right] \right] \\
&= \frac{1}{n} \mathbb{E} \left[\mathbb{E}_Q \left[\left| I_{\text{WSD}(P; \mathbf{P}_n) \leq \text{WSD}(Q; \mathbf{P}_n)} - I_{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})} \right| \middle| P_1, \dots, P_n \right] \right] \\
&\leq \frac{4C}{n} \mathbb{E} \left[\sup_{A \in \text{supp}(\mathbf{P}) \cup \text{supp}(\mathbf{Q})} |\text{WSD}(A; \mathbf{P}_n) - \text{WSD}(A; \mathbf{P})| \right] \\
&= O(n^{-3/2}),
\end{aligned}$$

318 where the last inequality is due to

$$\begin{aligned}
& \left| I_{\text{WSD}(P; \mathbf{P}_n) \leq \text{WSD}(Q; \mathbf{P}_n)} - I_{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})} \right| \leq \\
& I_{|\text{WSD}(P; \mathbf{P}) - \text{WSD}(Q; \mathbf{P})| \leq 2 \sup_{A \in \text{supp}(\mathbf{P}) \cup \text{supp}(\mathbf{Q})} |\text{WSD}(A; \mathbf{P}_n) - \text{WSD}(A; \mathbf{P})|}.
\end{aligned}$$

319 Then it follows Markov's inequality that $T_{11} = O_p(n^{-\alpha/4-1/2})$. By assumption (13), we bound T_{12} ,

$$\begin{aligned}
T_{12} &= \int \int I_{\text{WSD}(P; \mathbf{P}_n) \leq \text{WSD}(Q; \mathbf{P}_n)} d\mathbf{P} d\mathbf{Q} - \int \int I_{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})} d\mathbf{P} d\mathbf{Q} \\
&= \frac{1}{2} \int \int \left(I_{\text{WSD}(P; \mathbf{P}_n) \leq \text{WSD}(Q; \mathbf{P}_n)} + I_{\text{WSD}(P; \mathbf{P}_n) \geq \text{WSD}(Q; \mathbf{P}_n)} \right) d\mathbf{P} d\mathbf{Q} - \frac{1}{2} \\
&= \frac{1}{2} \int \int I_{\text{WSD}(P; \mathbf{P}_n) = \text{WSD}(Q; \mathbf{P}_n)} d\mathbf{P} d\mathbf{Q} \\
&= o(n^{-1/2}).
\end{aligned}$$

320 By assumptions (10) and (11), T_{13} is bounded by

$$\begin{aligned}
& \int \int \left(I_{\text{WSD}(P; \mathbf{P}_n) \leq \text{WSD}(Q; \mathbf{P}_n)} - I_{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})} \right) d(\mathbf{P}_n - \mathbf{P}) d\mathbf{Q} \\
& \leq \int \mathbb{P}_{Q \sim \mathbf{Q}} \{ \text{WSD}(P; \mathbf{P}_n) - o(1) \leq \text{WSD}(Q; \mathbf{P}) < \text{WSD}(P; \mathbf{P}) \} d(\mathbf{P}_n - \mathbf{P}) \\
& \leq C \int |\text{WSD}(P; \mathbf{P}_n) - o(1) - \text{WSD}(P; \mathbf{P})|^\alpha d(\mathbf{P}_n - \mathbf{P}) \\
& = o_p(n^{-1/2})
\end{aligned}$$

321 In summary, we showed that $\sqrt{n} T_1 = o_p(1)$ and $\sqrt{n} T_2 = o_p(1)$, and hence the claim follows, *i.e.*,

$$D(\mathbf{Q}_n, \mathbf{P}_n) - D(\mathbf{Q}_n, \mathbf{P}) = \int \int I_{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})} d\mathbf{Q} d(\mathbf{P}_n - \mathbf{P}) + o_p(n^{-1/2}).$$

322 By the central limit theory for empirical distribution process and Theorem 6.1 in [Bachoc et al., 2024],
323 we also obtain that

$$\sqrt{n} \int \int I_{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})} d\mathbf{Q} d(\mathbf{P}_n - \mathbf{P}) \xrightarrow{d} N(0, \sigma_{\mathbf{Q}\mathbf{P}}^2).$$

324 Therefore, it holds

$$D(\mathbf{Q}_n, \mathbf{P}_n) - D(\mathbf{Q}_n, \mathbf{P}) \xrightarrow{d} N(0, \sigma_{\mathbf{Q}\mathbf{P}}^2).$$

325 Following similar steps, we can also show that

$$\begin{aligned}
\sqrt{n} (D(\mathbf{Q}_n, \mathbf{P}) - D(\mathbf{Q}, \mathbf{P})) &= \sqrt{n} \int \int I_{\text{WSD}(P; \mathbf{P}) \leq \text{WSD}(Q; \mathbf{P})} d\mathbf{P} d(\mathbf{Q}_n - \mathbf{Q}) + o_p(1) \\
&\xrightarrow{d} N(0, \sigma_{\mathbf{P}\mathbf{Q}}^2).
\end{aligned}$$

326 Finally, we have

$$D(\mathbf{Q}_n, \mathbf{P}_n) - D(\mathbf{Q}, \mathbf{P}) = (D(\mathbf{Q}_n, \mathbf{P}_n) - D(\mathbf{Q}_n, \mathbf{P})) + (D(\mathbf{Q}_n, \mathbf{P}) - D(\mathbf{Q}, \mathbf{P})).$$

327 By the independence between $\{P_i\}_{i=1}^n$ and $\{Q_k\}_{k=1}^n$ and central limit theorem, the result follows. $\square$

328 C Additional details for numerical simulations

329 C.1 Case 1

330 The reference batches (distributed as $\mathbf{P}$) are drawn as

$$c_i \sim \text{Triangular}(-1, 0, 1), \quad X_j^i \mid c_i \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(c_i - 1, c_i + 1), \quad j = 1, \dots, 50.$$

331 The normal batches within the data stream are drawn in the same way. The abnormal batches
332 (distributed as $\mathbf{Q}$) are drawn from a discrete distribution supported on 200 equidistant points within
333 the interval $[-2, 2]$. Denoting the 200 points as $\{s_1, s_2, \dots, s_{200}\}$, the masses on these points are

$$\pi_k \propto \exp\left(-\frac{s_k^2}{2}\right), \quad \sum_{k=1}^{200} \pi_k = 1.$$

334 Each abnormal batch has 50 data points *i.i.d.* drawn from the above distribution.

335 The WSD-based p-values of regular batches are approximately uniform, with chi-squared goodness-
336 of-fit p-value being 0.238. The WSD-based p-values of abnormal batches are concentrated on $[0, 0.2]$
337 with chi-squared goodness-of-fit p-value being 1.53×10^{-40} .

338 C.2 Case 2

339 The reference and regular batches $P_i, i = 1, 2, \dots$ are drawn as follows: draw $r_i \sim \text{Unif}[0.02, 1]$;
 340 draw independent data points from $P_i = \text{Unif}[A_i]$ where A_i is a two-dimensional annulus with inner
 341 radii r_i and fixed area $\pi(1^2 - 0.02^2)$. That is, each P_i is a uniform distribution on a two-dimensional
 342 annulus with inner radii $r_i \sim \text{Unif}[0.02, 1]$ and fixed area.

343 The abnormal batches (distributed as $\mathbf{Q}$) are drawn as follows: drawn random points of coordinates
 344 $(r \cos \omega, r \sin \omega)$ with $\omega \sim \text{Unif}[(0, 2\pi], r \sim \sqrt{2} \text{Beta}(4, 3)$. The radius r is supported on $[0, \sqrt{2}]$
 345 with mean $4\sqrt{2}/7 \approx 0.808$ and a density peaked near 0.96, so the point cloud of $\mathbf{Q}$ forms a isotropic
 346 ring. Each abnormal batch has 50 data points *i.i.d.* drawn from the above distribution.

347 The WSD-based p-values of regular batches are approximately uniform, with chi-squared goodness-
 348 of-fit p-value being 0.416. The WSD-based p-values of abnormal batches are concentrated on $[0, 0.2]$
 349 with chi-squared goodness-of-fit p-value being 2.47×10^{-38} .

350 C.3 Additional details for anomaly detection in online regression

351 In practice, the effective batches are the gradient batches $\{\nabla \ell(\theta; X_i^t)\}_{i=1}^m$, as parameter updates
 352 depend only on gradients. Accordingly, in our experiments, $\hat{P}_t$ denotes the empirical distribution of
 353 the t -th batch of gradients. In online regression, we use squared norm loss

$$\ell_i^t(\theta) = \frac{1}{2} (y_i^t - \langle x_i^t, \theta \rangle)^2,$$

354 the per-sample residual gradient is

$$g_i^t(\theta) = \nabla_{\theta} \ell_i^t(\theta) = (\langle x_i^t, \theta \rangle - y_i^t) x_i^t.$$

355 Each batch is represented as the empirical measure

$$\hat{P}_t = \frac{1}{m} \sum_{i=1}^m \delta_{g_i^t}.$$

356 Detecting abnormal batches is equivalent to detecting the outlier $\hat{P}_t$. For fair detection, we compute
 357 WSD using gradient batches evaluated at a fixed anchor parameter θ_0 , namely $\{\nabla \ell(\theta_0; X_i^t)\}_{i=1}^m$,
 358 while the actual parameter updates still use the instantaneous gradients $\nabla \ell(\theta_t; X_i^t)$.

359 The regression parameter is updated by a Polyak average between the current estimate and the
 360 closed-form ordinary least squares solution on the current clean batch. Let

$$\tilde{\theta}_t = ((X^t)^\top X^t)^{-1} (X^t)^\top y^t,$$

361 where $X^t \in \mathbb{R}^{m \times d}$ has $\{x_i^t\}_{i=1}^m$ as rows and $y^t \in \mathbb{R}^m$ stacks $\{y_i^t\}_{i=1}^m$. The online update is

$$\hat{\theta}_{t+1} = (1 - \alpha) \hat{\theta}_t + \alpha \tilde{\theta}_t,$$

362 with step size $\alpha = 0.02$.

363 D Additional details for real data analysis

364 D.1 MNIST stream construction

365 The real-data experiment uses the MNIST data set of handwritten digits. Every image is cut into
 366 28×28 pixels. At each t , $m = 60$ images are drawn from the ten digits $\{0, 1, \dots, 9\}$ with proportions

$$\pi_t = (1 - \beta_t) \pi_0 + \beta_t \pi_1, \quad \beta_t = \frac{t}{T-1}, \quad t = 0, 1, \dots, T-1,$$

367 where π_0 and π_1 are sampled from a symmetric Dirichlet distribution with concentration parameter
 368 $c = 0.4$ of order 10. Anomalous batches are produced by randomly mis-labelling a fraction 10% of
 369 the images within this batch.

D.2 Classifier built on convolutional neural network

The classifier is built upon a convolutional neural network (CNN). It maps the input image to a 64-dimensional latent feature vector and then applies a linear classification head with 10 output logits. The classifier is trained online by SGD with cross-entropy loss; we run 5 inner gradient steps per incoming batch with learning rate 0.05.

In this analysis, we use cross-entropy loss. The gradient depends on both the data (x, y) and on the parameter θ . To compare the data batches $\{(x_i^t, y_i^t)\}_{i=1}^m$ for $t = 0, 1, \dots$ in a fair and effective manner, we fix an anchor θ_0 and evaluate gradients evaluated at θ_0 . This strategy is only used in computing WSDs. When updating parameter in SGD, we use the instantaneous gradient $\nabla \ell(\theta_t; (x_i^t, y_i^t))$. The gradient of the cross-entropy loss with respect to the logits of θ_0 ,

$$\nabla \ell(\theta_0; (x_i^t, y_i^t)) = p_i^t - e_{y_i^t}, \quad p_i^t = \text{softmax}(f_{\theta_0}(x_{t,i})),$$

where $e_{y_i^t} \in \mathbb{R}^{10}$ is the one-hot encoding of the observed label. The collection of gradients $\{\nabla \ell(\theta_0; (x_i^t, y_i^t))\}_{i=1}^m$ is treated as the $\hat{P}_t$.

E Computation details of existing detectors

We compare the WSD-based p -value against four classical anomaly detectors under the same settings. The four methods are based on

Gradient norm.

$$\text{score}_t^{\text{loss}} = \frac{1}{m} \sum_{i=1}^m \|g_i^t\|_2. \quad (17)$$

Mean-grad L_2 .

$$\text{score}_t^{\text{mg}} = \|\bar{g}_t\|_2. \quad (18)$$

Hotelling T^2 .

$$\text{score}_t^{T^2} = m(\bar{g}_t - \hat{\mu})^\top (\hat{\Sigma} + \epsilon I_{d \times d})^{-1} (\bar{g}_t - \hat{\mu}), \quad \epsilon = 10^{-6} \quad (19)$$

with ϵ introduced for numerical stability.

MMD with Gaussian RBF kernel.

$$\text{score}_t^{\text{MMD}} = \frac{1}{m^2} \sum_{i,j} k(g_i^t, g_j^t) + \frac{1}{L^2} \sum_{i,j} k(\tilde{g}_i, \tilde{g}_j) - \frac{2}{mL} \sum_{i,j} k(g_i^t, \tilde{g}_j), \quad (20)$$

where $\{\tilde{g}_j\}_{j=1}^L$ is a uniform sub-sample of size $L = 200$ from the reference pool, $k(u, v) = \exp(-\|u - v\|^2 / \sigma_k^2)$, and σ_k^2 is set by the median heuristic on the pooled distances $\{\|u - v\|^2\}$.

AUCs are computed by comparing each score against the ground-truth labels on the 95 evaluated rounds.